for two variables we will have two integrals.
that gives us volume under a surface

e.g Solve $\iint_R (x - 3y^2)\,dA$ where $R = \{(x,y) \mid 0 \le x \le 2,\ 1 \le y \le 2\}$

Solution $\iint_R$ → R means Region

Region

$$\int_0^2 \int_1^2 f(x,y)\,dy\,dx$$ first solve inner integral

$$\int_0^2 \int_1^2 x - 3y^2\,dy\,dx \rightarrow \int_0^2 \left| xy - \frac{3y^3}{3} \right|_1^2 dx$$

keep x constant
when integrating WRT y.

$$\int_0^2 \left| \left(2x - \frac{3(8)}{3}\right) - \left(x - \frac{3(1)}{3}\right) \right|_1^2 dx$$

$$\int_0^2 (2x-8) - (x-1)\,dx$$

$$\int_0^2 [2x-8-x+1]\,dx$$

$$\int_0^2 [x-7]\,dx \Rightarrow \left[\frac{x^2}{2} - 7x\right]_0^2$$

$$= \left[\frac{(2)^2}{2} - 7(2)\right] - \left[\frac{(0)^2}{2} - 7(0)\right]$$

$$= -12$$

Volume is

hence surface is below the axis.

Can we change orders of x and y limits.

$$\int_1^2 \int_0^2 x - 3y^2 \, dx \, dy = -12$$ hence we are getting same Answer.

we can Switch the limits

$$\int_a^b \int_c^d f(x,y) \, dy \, dx = \int_c^d \int_a^b f(x,y) \, dx \, dy$$

if limits are constants. and not in variable form.

e.g $\int_1^{x^2}$ or $\int_{-1}^{\sqrt{x}+y}$ cannot change order.

$$\int_1^2 \int_0^{x^2} f(x,y) \, dy \, dx \neq \int_0^{x^2} \int_1^2 f(x,y) \, dx \, dy$$

OR we can Switch limits if Region is Rectangle

e.g Evaluate the iterated integral.

i) $\int_0^3 \int_1^2 x^2 y \, dy \, dx$

Solving inner function by keeping x constant.

$$\int_0^3 \left| \frac{x^2 y^2}{2} \right|_1^2 dx$$

$$\int_0^3 \left[\left(\frac{x^2(4)}{2} \right) - \left(\frac{x^2(1)}{2} \right) \right] dx$$

$$\int_0^3 \frac{3}{2} x^2 \, dx \rightarrow \left| \frac{3x^3}{6} \right|_0^3 = \frac{27}{2}$$

↓

Volume

ii) $\int_1^2 \int_0^3 x^2 y \, dx \, dy$

Solving inner integral by keeping y constant.

$$= \int_1^2 \left| \frac{x^3 y}{3} \right|_0^3 dy$$

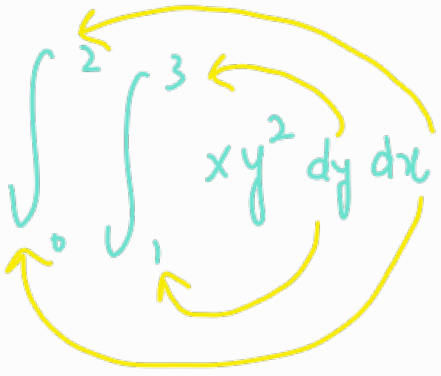
$= \int_1^2 4y\,dy \rightarrow \left|_1^2 \frac{4y^2}{2}\right| \rightarrow \frac{a^2}{2}$

↓

Volume

Double Integrals example:. / Iterated Integrals:

1)



Always integrate from inside to outside

$$\int_1^3 xy^2\,dy = \left|_1^3 \frac{xy^3}{3}\right| = \left|\frac{27x}{3} - \frac{x}{3}\right)$$

$$= \boxed{\frac{26}{3}x}$$

$$\int_0^2 \frac{26}{3}x\,dx = \left|_0^2 \frac{26x^2}{3\cdot 2}\right| = \boxed{\frac{52}{3}}$$

e.g $\int_0^1 \int_0^{2y} [4 + 2x - y^2]\,dx\,dy$

$$\Rightarrow \int_0^{2y} 4 + 2x - y^2 \, dx \Rightarrow \left| 4x + \frac{2x^2}{2} - y^2 x \right|_0^{2y}$$

$$\Rightarrow 4(2y) + \frac{2(2y)^2}{2} - y^2(2y)$$

$$\Rightarrow \boxed{8y + 4y^2 - 2y^3}$$

$$\int_0^1 8y + 4y^2 - 2y^3 \, dy \Rightarrow \left| \frac{8y^2}{2} + \frac{4y^3}{3} - \frac{2y^4}{4} \right|_0^1$$

$$\Rightarrow \left| 4y^2 + \frac{4}{3}y^3 - \frac{y^4}{2} \right|_0^1 \Rightarrow \boxed{\frac{23}{6}}$$

(finding limits in double integrals)

↓

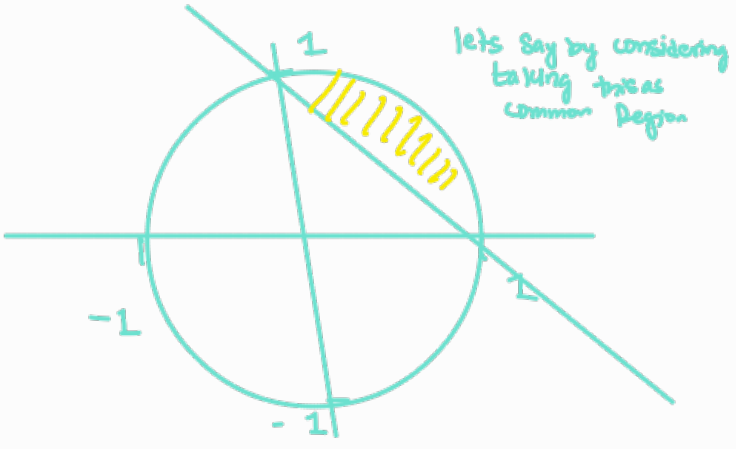
when they are not constant. / when Region R is general or not Rectangle.

e.g

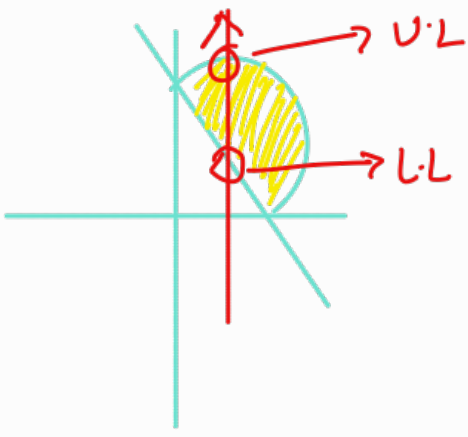
R is given $x^2+y^2=1$ > $x+y=1$ $y=-x+1$

1) Make the Region By sketching it

$$\iint_R f(x,y)\,dy\,dx = \int_{?}^{?}\int_{?}^{?} f(x,y)\,dy\,dx$$



2) Take a vertical line in increasing direction of y. (this is for y-limit)



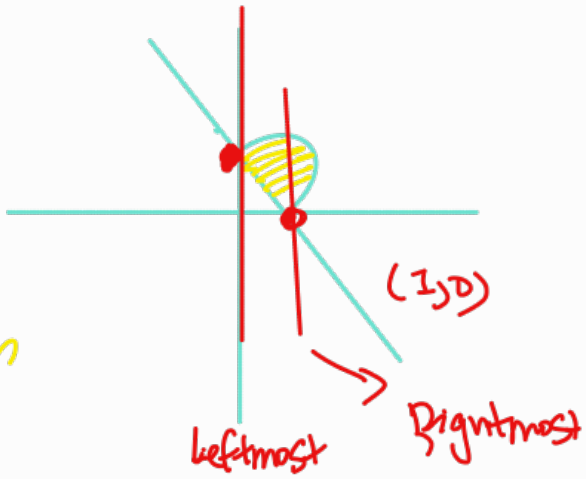
from where the line enters the Region is its lower limit and from where the line exists the Region is the upper limit

Entrance $x+y=1$ $\boxed{y=1-x}$

exit $x^2+y^2=1$ $y^2=1-x^2$

y-Limits ← $\boxed{y=\sqrt{1-x^2}}$

$$\int_{\sqrt{1-x^2}}^{1-x} f(x,y)\,dy\,dx$$



3) x limits of integration

Start with the left most vertical line. $x=0$ that will be the lower limit.

note:: These lines are common points of Both curves (and).

Higher limit $x=1$

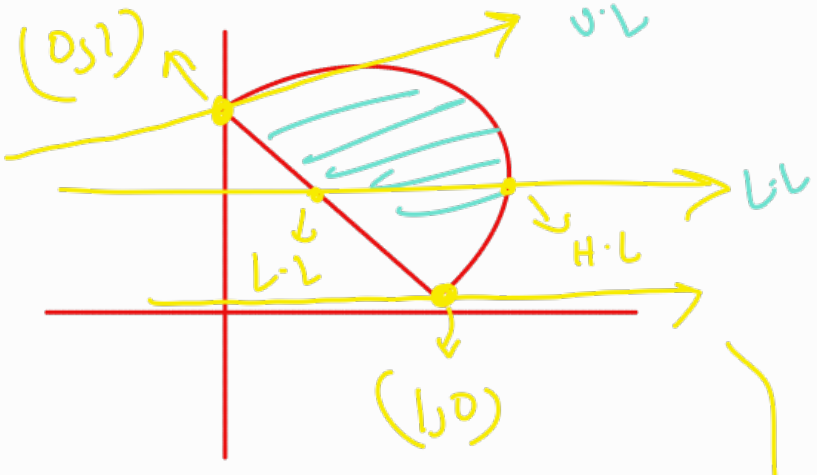
$$\int_0^1 \int_{\sqrt{1-x^2}}^{1-x} f(x,y)\,dy\,dx$$

Now find limits for

$$\int_{?}^{?}\int_{?}^{?} f(x,y)\,dx\,dy = ?$$

Always find Inner limits first

Same steps.



Step 1 Sketch the Region

Step 2:: find x-limits

By making a horizontal line in Increasing x-direction

(H.L $x^2+y^2=1$

$x^2=1-y^2$

$x=\sqrt{1-y^2}$

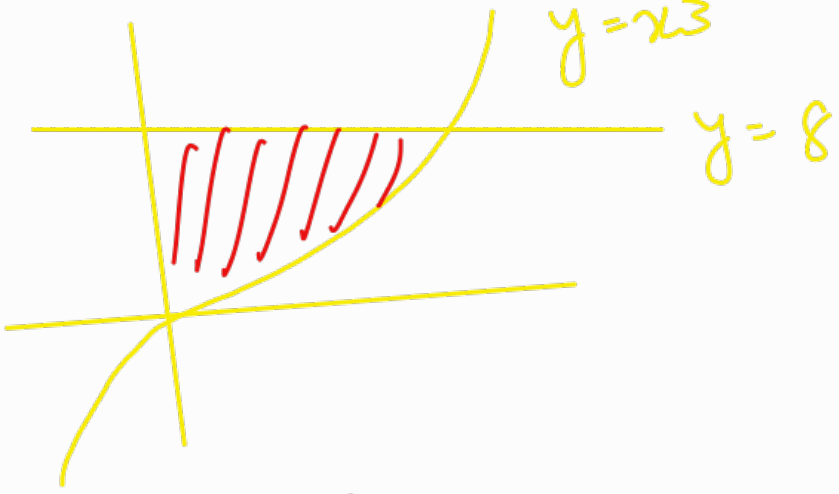
L.L $x+y=1$ $\quad x=1-y$

Step 3

For y-limits find the upper most horizontal line that passes through Both curves that will be the U.L and the lowest horizontal like that passes through Both curves will be L.L

$$\int_0^1 \int_{1-y}^{\sqrt{1-y^2}} f(x,y)\,dx\,dy$$

e.g



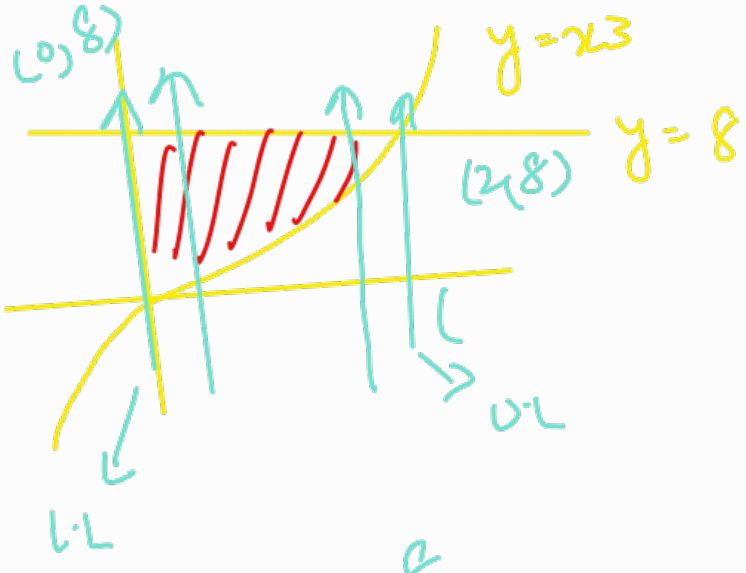
note:: Region will be in x-y plane.

find Limit $\iint f(x,y)\,dy\,dx$.

first find y-limits.

L.L $y = x3$

U.L $y = 8$



Now find x-limits

Use Vertical

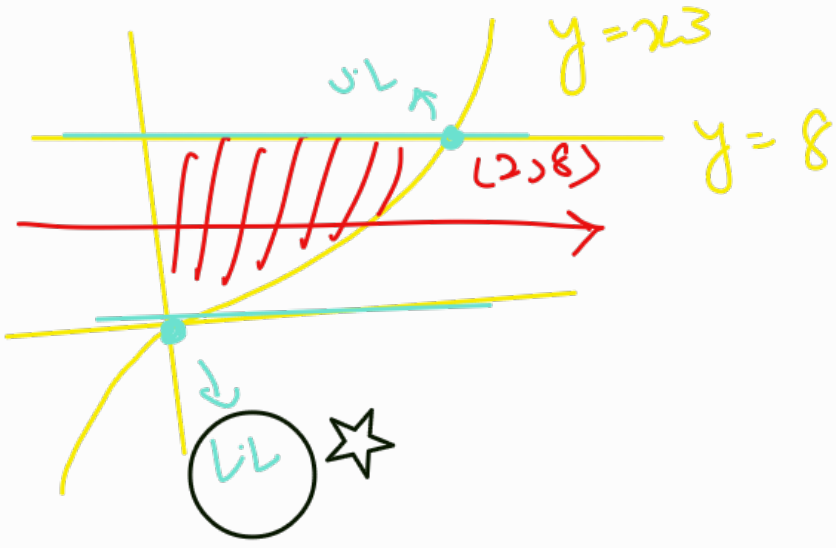
leftmost $x = 0$

and Rightmost

$x = 2$

$$\int_0^2 \int_{x3}^{8} f(x,y)\,dy\,dx$$

now using x-limits first



$$\iint f(x,y)\,dx\,dy$$

1) Now using horizontal cross-section.

L.L = $x = 0$

U.L = $y = x^3$ $x = y^{1/3}$

a) y-limits

U.L $y = 8$

L.L = $y = 0$

$$\int_{x=0}^{y^{1/3}} \int f(x,y)\,dx\,dy$$

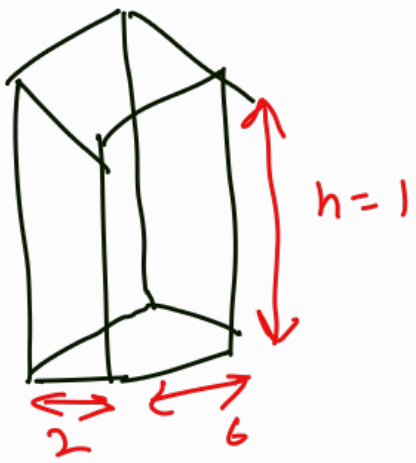
Area by double integrals

if $f(x,y) = 1$ $z = 1$

$$\int_a^b \int_c^d 1\,dy\,dx = \text{Area of region}$$

(L) (L)

e.g



Vol = $1 \times 2 \times 6 = 12$

Area = $6 \times 2 = 12$

hence if one side is 1 it gives us Area.

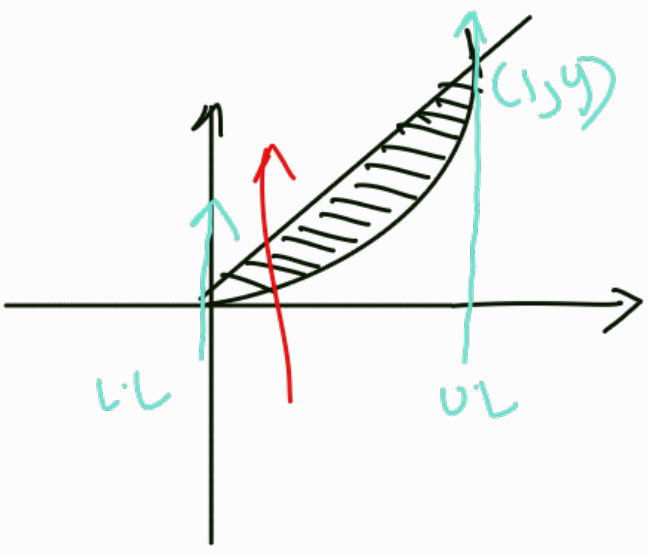
e.g

$$A = \iint_R (1)\, dy\,dx \quad \text{OR} \quad \iint_R dA$$

Q1

Find the area of the Region R bounded by $y = x$ and $y = x^2$ in first Quadrant.

to find area $f(x,y)$ must be 1



$$A = \iint_R dA = \iint dy\,dx$$

finding limits- y-limits first.

L.L $y = x^2$ U.L $y = x$

finding limits x-limits

L.L $x = 0$ U.L $x = 1$ $y = x^2$ and $y = x$

$x^2 = x$

$x^2 - x = 0$

$x(x-1) = 0$

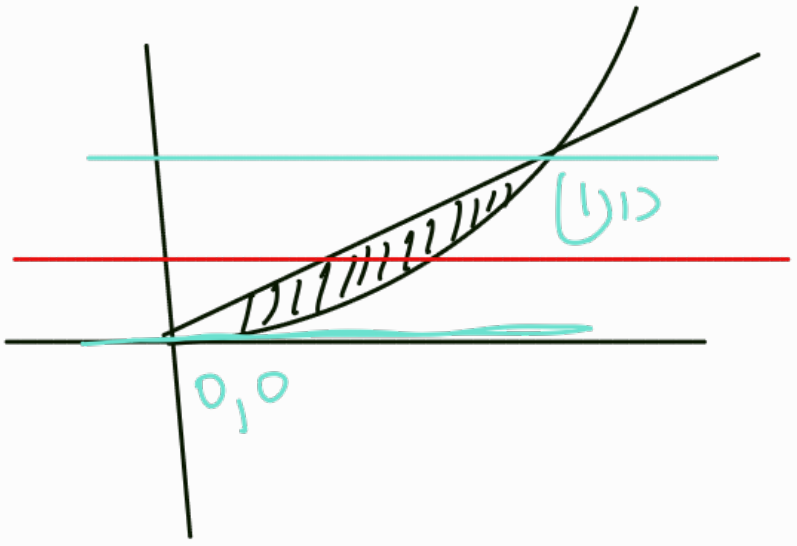
$x = 0$ $x = 1$

$$\int_0^1 \int_{x^2}^{x} 1 \, dy\,dx = \int_0^1 \Big| y \Big|_{x^2}^{x} \, dx$$

$$= \int_0^1 x - x^2 \, dx$$

$$= \Big| \frac{x^2}{2} - \frac{x^3}{3} \Big|_0^1 = \boxed{\frac{1}{6}}$$

now Solving with x-limits first.



U.L $y = x^2$

$\boxed{x = \sqrt{y}}$

L.L $y = x$

$\boxed{x = y}$

finding y-limits

$y = 0 \rightarrow$ L.L $\quad (y = 1) \rightarrow$ U.L

$y = x \quad y = x^2$

$x = x^2$

$x^2 - x = 0$

$x(x-1) = 0$

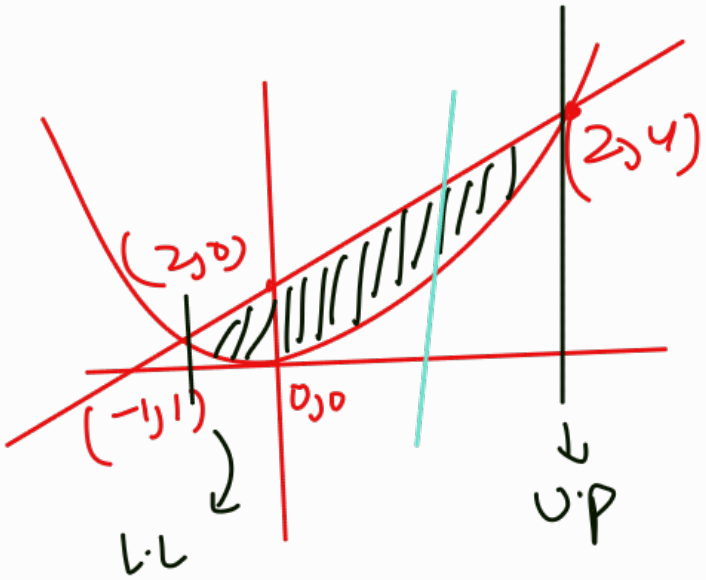
$x = 0, x = 1$

$$\int_0^1 \int_y^{\sqrt{y}} 1 \, dx\,dy = \int_0^1 \int_y^{\sqrt{y}} x \, dy$$

$$= \int_0^1 \left(\sqrt{y} - y\right) dy = \left| \frac{2y^{3/2}}{3} - \frac{y^2}{2} \right|_0^1$$

$$= \boxed{\frac{1}{6}}$$

example: find the area of Region R enclosed by $y = x^2$ and $y = x + 2$



finding y-limits

$y = x^2$ - L.L

$y = x + 2$ - U.L

finding x-limits

$x = -1$ and $x = 2$

$$\int_{-1}^{2}\int_{x^2}^{x+2} 1 \, dy\,dx = \int_{-1}^{2} \Big| y \Big|_{x^2}^{x+2} dx$$

$$= \int_{-1}^{2} (x+2) - (x)^2 \, dx$$

$$= \int_{-1}^{2} x+2-x^2 \, dx$$

$$= \Big| \frac{x^2}{2} + 2x - \frac{x^3}{3} \Big|_{-1}^{2} = \frac{10}{3} - \left(\frac{-7}{6}\right) = \frac{9}{2}$$

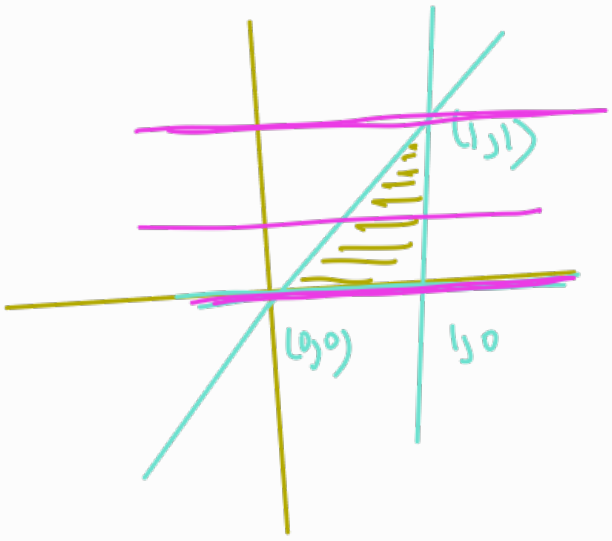
Ex 15.2 ex 2

$$\iint_R \frac{\sin x}{x}\, dA$$

R :: is a triangle

$x = 1$ ⟩ x-axis ⟩

$y = x$



x-limits

$y = x$ - L.L

$x = 1$ - U.L

y-limits

$y = 0$ L.L

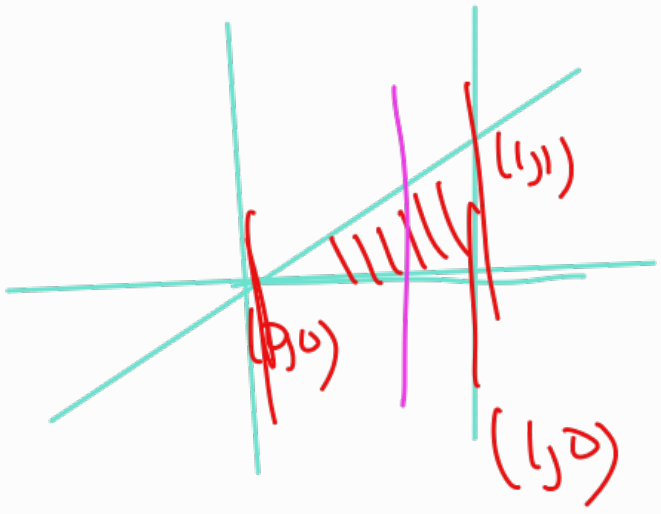
$y = 1$ U.L

$$\iint f(x,y)\, dx\, dy$$

$$\int_0^1 \int_y^1 \frac{\sin x}{x}\, dx\, dy$$

$$\int_0^1 \int_y^1 \frac{\sin x}{x}\, dx\, dy \rightarrow$$

Cannot integrate hence try other order



y-limits

$y=0 \rightarrow$ L.L

$y=x \rightarrow$ U.L

x-limits

$x=0$ – L.L

$x=1$ – U.L

$$\int_0^1 \int_0^x \frac{\sin x}{x}\, dy\, dx$$

$$\int_0^1 \left| \int_0^x y \frac{\sin x}{x} \right| dx \rightarrow \int_0^1 (\sin x)\, dx$$

$\sin x \rightarrow \cos$

$\cos 2 - \sin x$

$$\Rightarrow \left| \int_0^1 -\cos x \right| \Rightarrow -\cos(1) - (-\cos(0))$$

$\rightarrow$

$$\int_0^2 \int_{x^2}^{2x} 4x+2 \, dy \, dx$$

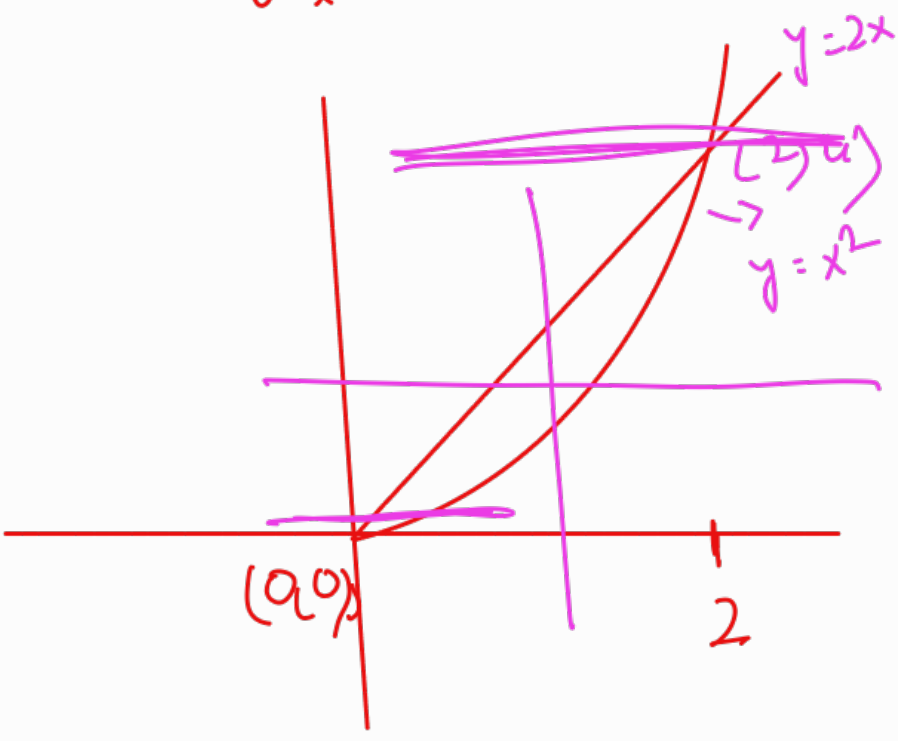
graph

$y = 2x$ — U.L

$y = x^2$ — L.L

$x = 2$ — U.L

$x = 0$ — L.L



Reserved order

L.L $x \rightarrow y = 2x$ $\boxed{\frac{y}{2} = x}$

U.L $x \rightarrow y = x^2$ $\boxed{x = \sqrt{y}}$

L.L $y \rightarrow y = 0$

U.L $y \rightarrow y = 4$

$$\int_0^4 \int_{y/2}^{\sqrt{y}} (4x+2)\,dx\,dy$$

$$\int_0^4 \int_{y/2}^{\sqrt{y}} \frac{4x^2}{2} + 2x \, dy \rightarrow \int_0^4 \left(\frac{4(\sqrt{y})^2}{2} + 2(\sqrt{y}) \right) - \left(\frac{4\left(\frac{y}{2}\right)^2}{2} + 2\left(\frac{y}{2}\right) \right)$$

$$\rightarrow \int_0^4 \left(\frac{4y}{2} + 2\sqrt{y} - \frac{4y^2}{8} - y \right)$$

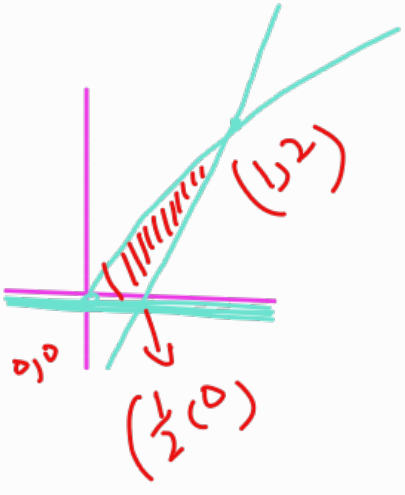
$$\rightarrow \int_0^4 y + 2\sqrt{y} - \frac{4y^2}{8}$$

$$\rightarrow \left| \frac{y^2}{2} + \frac{4y^{3/2}}{3} - \frac{4y^3}{24} \right|_0^4$$



$z = 16 - x^2 - y^2$

Region :- $y = 2\sqrt{x}$ $\quad y = 4x - 2$ $\quad x = 0$



using y-limits

$$\iint f(x,y)\, dy\, dx$$

L.L $\quad x = 0$

U.L $\quad y = 2\sqrt{x}$

X-limits

$x = 0$

$x = 1$

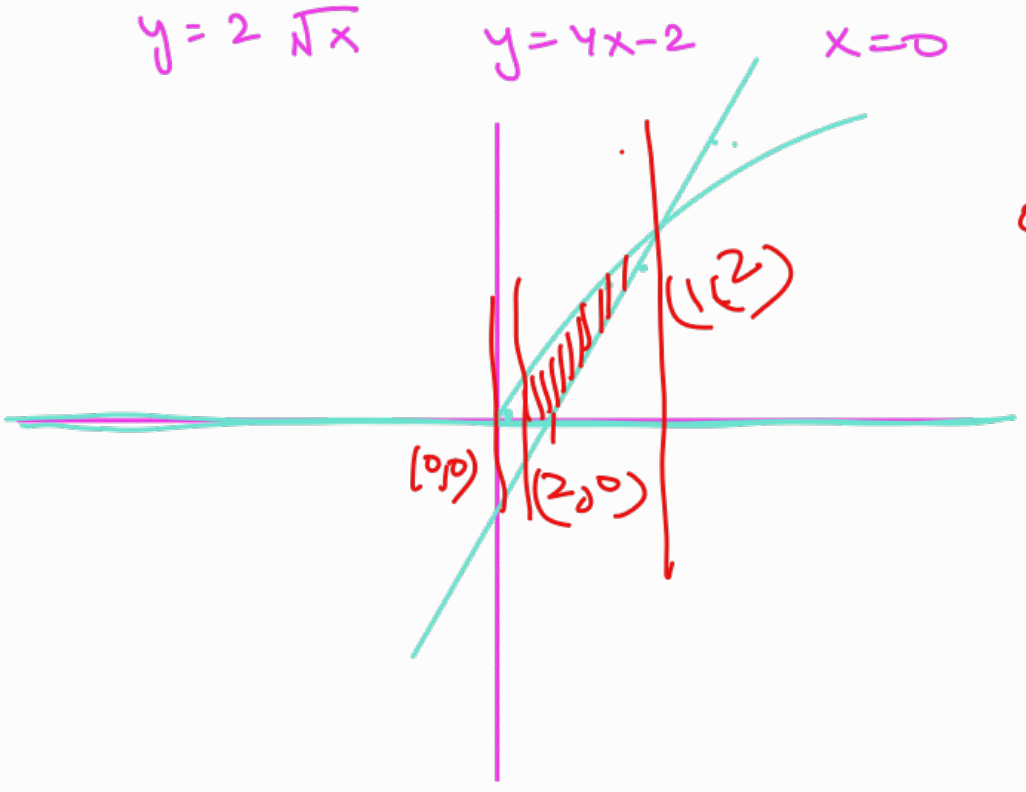
$$\int_0^1 \int_0^{2\sqrt{x}} 16 - x^2 - y^2 \, dy\, dx$$

$$\int_0^1 \int_0^{2\sqrt{x}} \left| 16y - x^2 y - \frac{y^3}{3} \right| dx$$

$2\sqrt{x}$

$$\int_0^1 \left(32\sqrt{x} - 2\sqrt{x} \cdot x^2 - \frac{8x^{3/2}}{3} \right) dx$$

$$\int_0^1 \left(\frac{32x^{3/2}}{3/2} - \frac{2x^{7/2}}{7/2} - \frac{8x^{5/2}}{3 \times \frac{5}{2}} \right)$$



x	0	1	2
0	2	2.8	

$0 = 4x - 2$

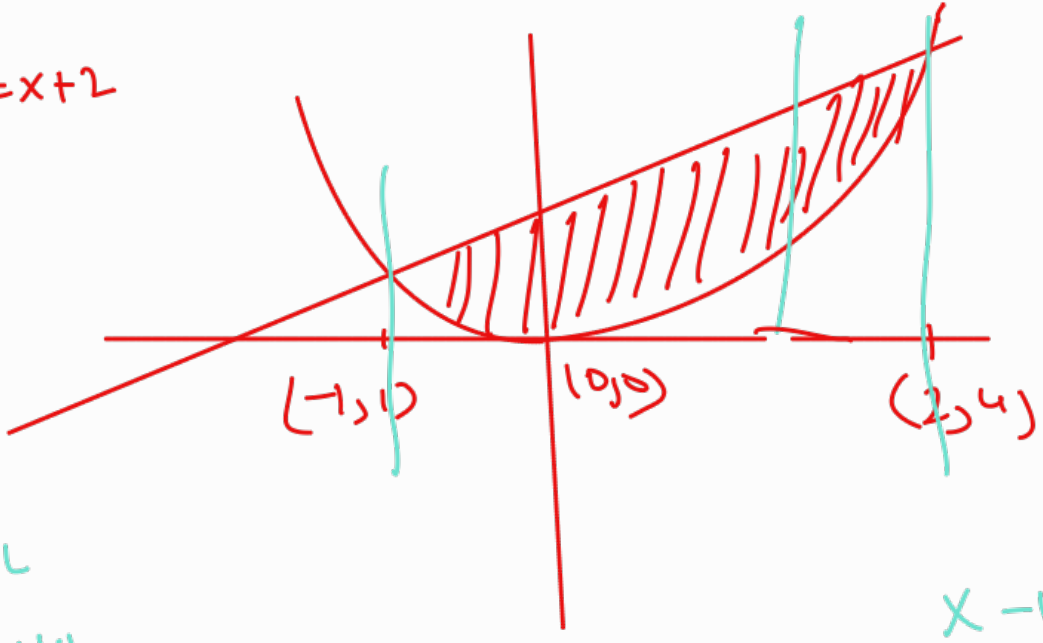
$2 = 4x$

$x = 2$

$y = 4x - 2$

$y = x^2$ $y = x+2$

$y = x+2$



y-limits

$y = x^2 \rightarrow$ L.L

$y = x+2 \rightarrow$ U.L

X-limits

$x = -1 \rightarrow$ L.L

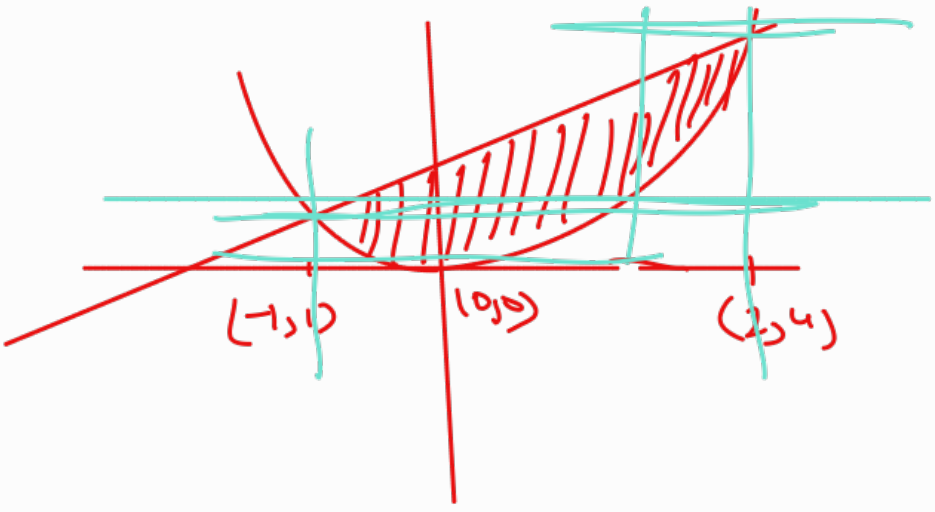
$x = 2 \rightarrow$ U.L

$$\int_{-1}^{2} \int_{x^2}^{x+2} 1 \, dy \, dx$$

$$\int_{-1}^{2} \int_{x^2}^{x+2} y \, dy \, dx \qquad \int_{-1}^{2} (x+2) - (x^2) \, dx$$

$$\left| \frac{x^2}{2} + 2x - \frac{x^3}{3} \right|_{-1}^{2} \rightarrow \frac{10}{3} - \left(-\frac{7}{6}\right)$$

$$\rightarrow \frac{9}{2}$$



x - limits

L.L $y = x+2$

$x = y-2$

U.L $y = x^2$

$x = \pm\sqrt{y}$

y - limits

$y = 4 \rightarrow$ U.L

$x = 0 \rightarrow$ L.L

$$\int_0^4 \int_{y-2}^{\sqrt{y}} 1 \, dx\,dy$$

$$\int_0^4 \left| x \right|_{y-2}^{\sqrt{y}} dy \rightarrow \int_0^4 \sqrt{y} - (y-2)\, dy$$

$$\int_1^4 (\sqrt{y} - y + 2)\, dy \Rightarrow \left| \frac{y^{\frac{3}{2}}}{\frac{3}{2}} - \frac{y^2}{2} + 2y \right|_0^4$$

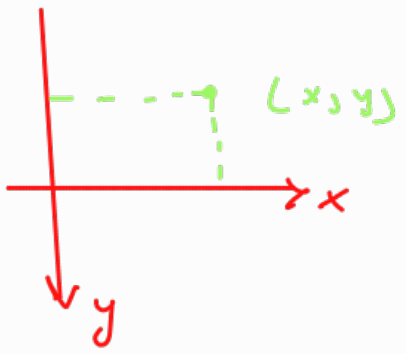
$$\frac{16}{3} - \frac{12}{6}$$

Ch: 15.4

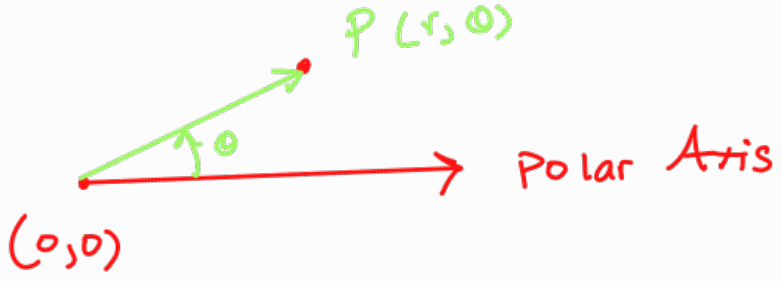
Introduction to polar coordinates. Converting double integral to polar coordinate

$$\iint f(x,y)\,dy\,dx = \iint f(r,\theta)\,r\,dr\,d\theta$$

(Cartesian co ordinates)

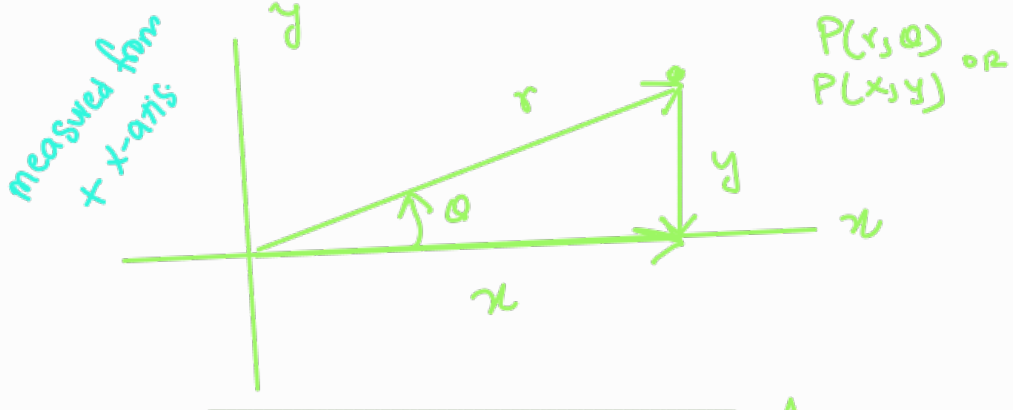


(Polar co ordinates)



☆ we will draw a ray from the origin. The length of the ray has a magnitude (r)

r, θ is called Polar co ordinates



P(r, θ) or P(x, y)

P(r, θ)
length Angle

$y = \text{Sin}\theta r$
$x = \cos\theta r$ ☆

$\sin\theta = \frac{y}{r}$

$\cos\theta = \frac{x}{r}$

To go from polar to calersian co ordinates we write: $x = r\cos\theta$, $y = r\sin\theta$

To go from calersion to polar co ordinates we write $x^2 + y^2 = r^2$ $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

e.g 1 find the rectangular co ordinates of the point

$Q\left(3, \frac{5\pi}{6}\right) = Q(r, \theta)$

↓ means the same thing as Cartesian co ordinates

going from Polar to Cartesian ↑

$x = r\cos\theta \qquad y = r\sin\theta$

$x = 3\cos\left(\frac{5\pi}{6}\right) \qquad y = 3\sin\left(\frac{5\pi}{6}\right)$

$x = -\frac{3\sqrt{3}}{2} \qquad y = \frac{3}{2}$

$$\left(-\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$$

e.g Find the Polar co ordinates of a point $P(3,2)$

C → P

$x^2 + y^2 = r^2 \qquad (3)^2 + (2)^2 = r^2$

$x + y = r^2$

$$r = \sqrt{13}$$

$\theta = \tan^{-1}\left(\frac{2}{3}\right)$

$\theta = \tan^{-1}\left(\frac{2}{3}\right)$

final Ans $\sqrt{13}, \tan^{-1}\left(\frac{2}{3}\right)$

Double Integrals in Polar form:

$$\iint_R f(x,y)\,dy\,dx = \iint_R f(r,\theta)\,r\,dr\,d\theta$$

↓ in Polar form.

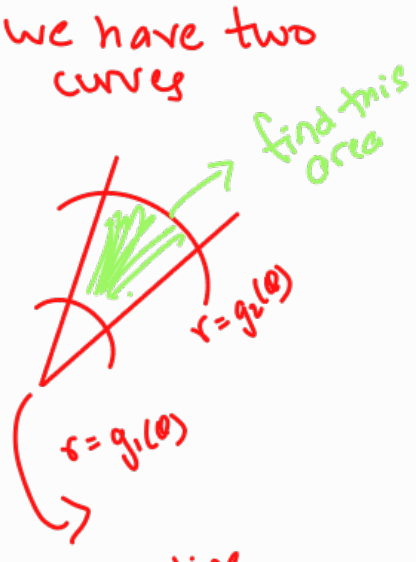
note we write double Integrals in Polar and triple Integral in Cylindrical co ordinates.

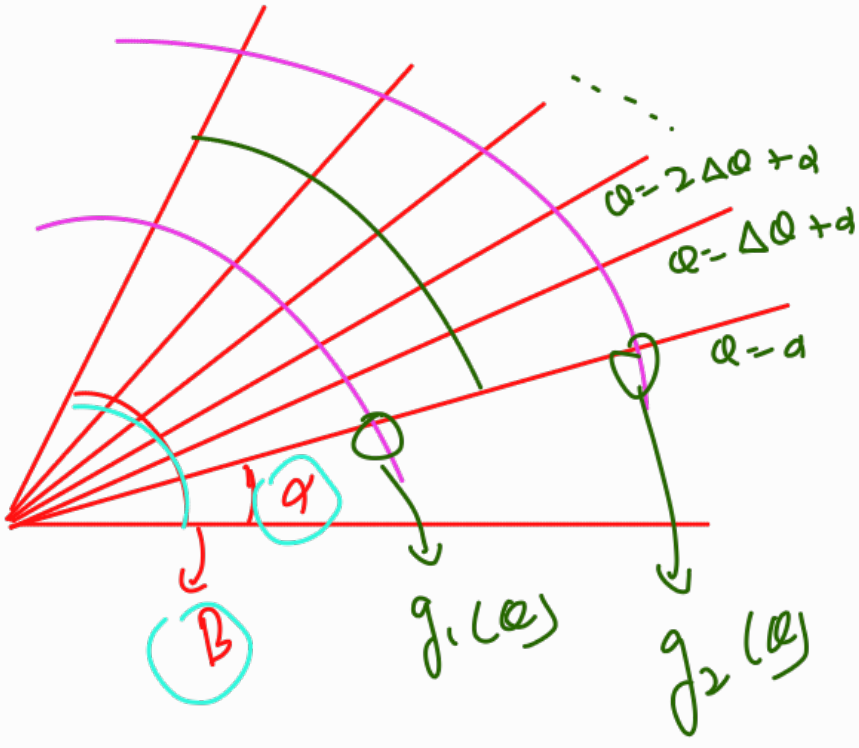
Triple Integral is written in cylindrical coordinate

$$\iiint_R f(x,y,z)\,dz\,dy\,dx = \iiint_R f(r,\theta,z)\,r\,dr\,d\theta\,dz$$

cylindrical coordinates

How to write a double integral in Polar form.



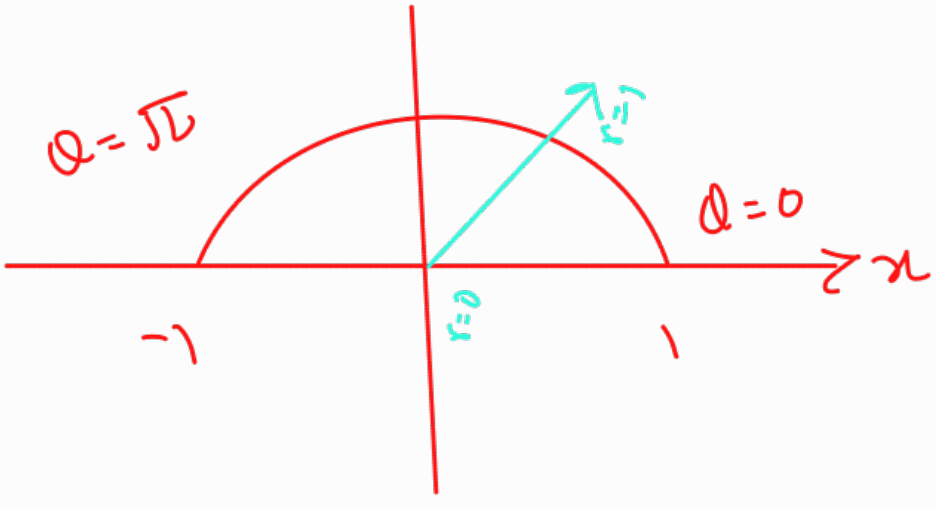


$$\iint_R f(x,y)\,dy\,dx = \iint_R f(r,\theta)\,dA = \int_{\theta=\alpha}^{\theta=\beta}\int_{r=g_1(\theta)}^{r=g_2(\theta)} f(r,\theta)\,r\,dr\,d\theta$$

(Lower curve)

e.g Evaluate $\iint_R e^{x^2+y^2}\,dy\,dx$

where R is the Semi-circle Region bounded by the x-axis and the curve $y=\sqrt{1-x^2}$



$y = r\sin\theta$

$x = r\cos\theta$

Transforming $y=\sqrt{1-x^2}$

$$r\sin\theta = \sqrt{1-(r^2\cos^2\theta)}$$

$$r^2\sin^2\theta = 1 - r^2\cos^2\theta$$

$$r^2\sin^2\theta + r^2\cos^2\theta = 1$$

$$r^2(1) = 1$$

$r^2 = 1$ $\boxed{r = 1}$

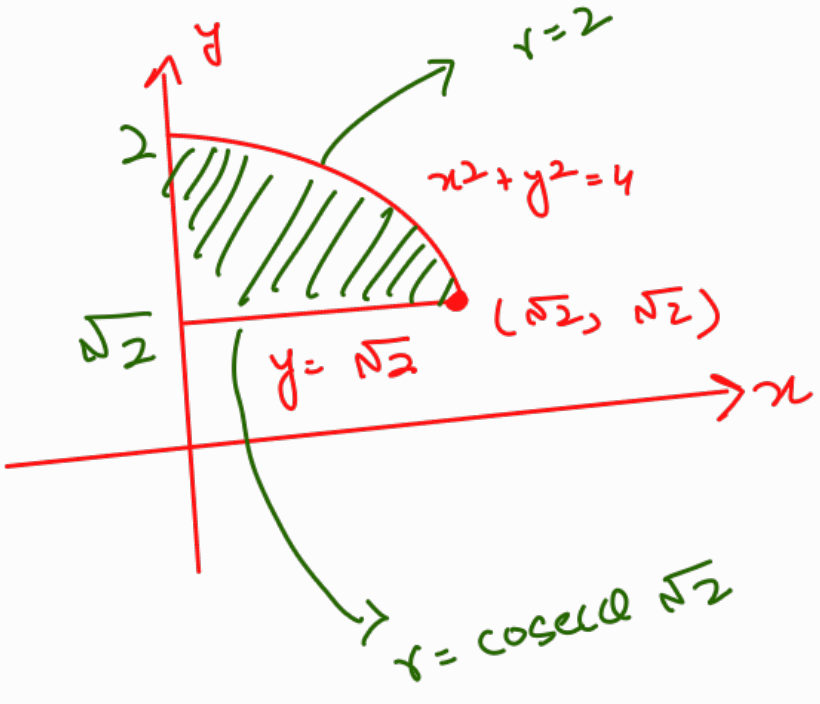
↓

Catersion to Polar co ordinates

$$\iint_R e^{x^2+y^2}\, dy\, dx$$ → convert into polor form

$$= \boxed{\int_{\theta=0}^{\theta=\pi}\int_{r=0}^{r=1} e^{r^2} r\, dr\, d\theta}$$

finding limits in polor co ordinates



$x^2+y^2 = r^2$

$r^2 = 4$

$\boxed{r = 2}$

$y = r\sin\theta$

$r\sin\theta = \sqrt{2}$

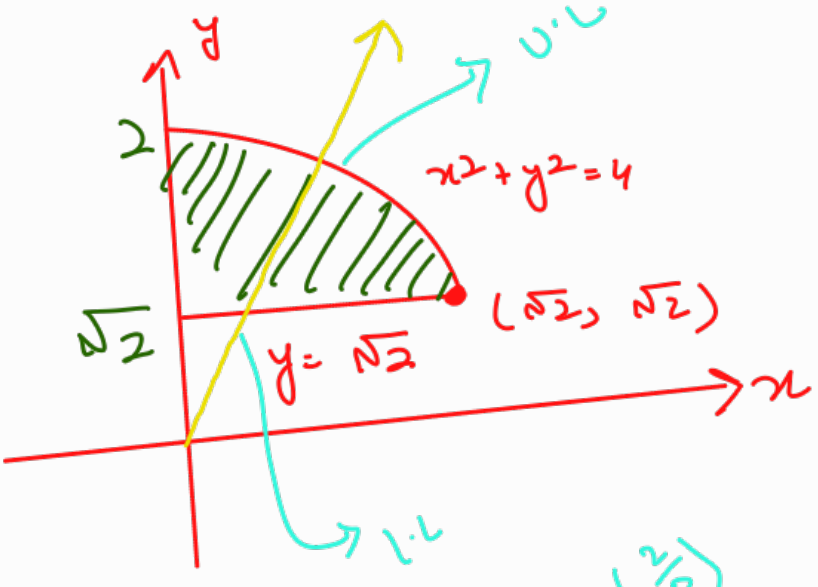
$r = \dfrac{\sqrt{2}}{\sin\theta}$

$$\int\int_{\text{cosec}\theta\sqrt{2}}^{2} f(r,\theta)\, r\, dr\, d\theta$$

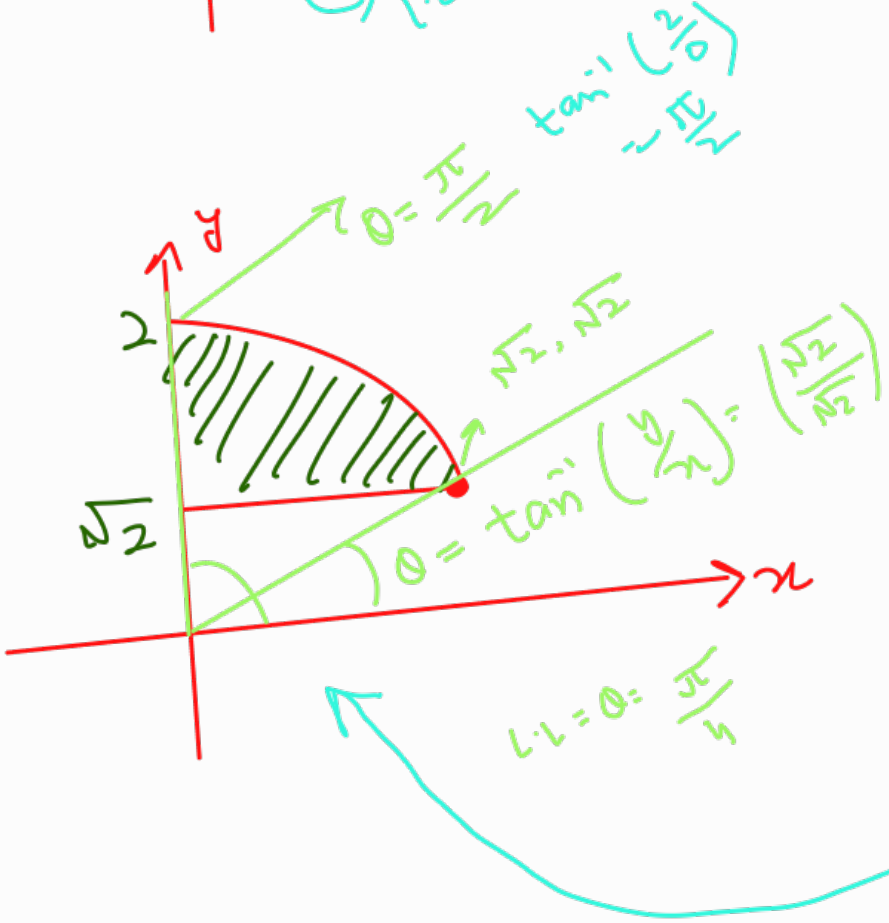
Step 1: Plot the Region / Sketch

Step 2: find r-limits of Integration. ↓

By drawing A ray from OP



Step 3: From Day enters → L.L and exists that is U.L



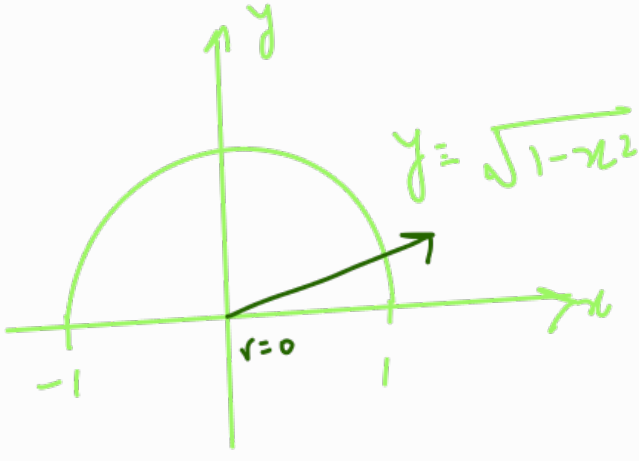
Step 4) find θ limits of Integration

By again drawing a ray from where the graph starts and where it ends.

$$\iint_R f(x,y)\,dy\,dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{r=\sqrt{2}\,\mathrm{cosec}\,\theta}^{r=a} f(r,\theta)\, r\,dr\,d\theta$$

e.g Evaluate

$$\iint_R e^{x^2+y^2}\,dy\,dx$$



first taking a ray to find r limits.

U.L of r is $y=\sqrt{1-x^2}$ in Calersian and in polar it is $r=1$ and Lower Limit is $r=0$

$$\int_0^{\pi} \int_0^{1} e^{r^2}\, r\,dr\,d\theta$$

$$= \frac{\pi}{2}(e-1)$$

Triple Integrals in Rectangular coordinates

$\int f(x)\,dx$ = Area under the curve

$\iint_R f(x,y)\,dx\,dy$ = Volume under the Surface

$\iiint_D f(x,y,z)\,dz\,dy\,dx$ → hyper Volume / Volume in 4D

(height: $f(x,y,z)$; Volume: $dz\,dy\,dx$)

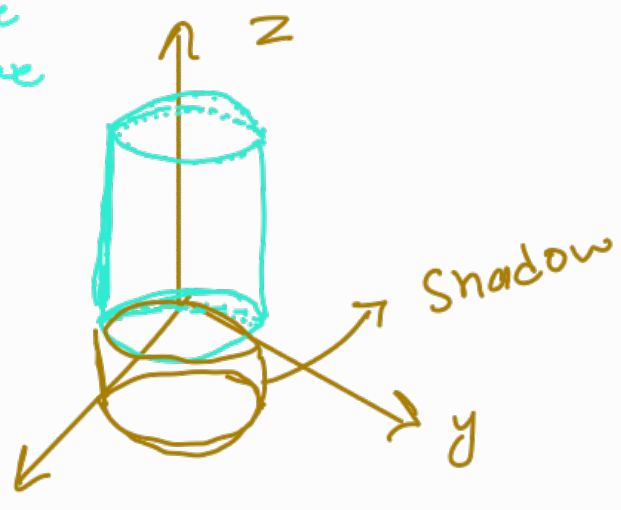
However lets say $f(x,y,z)=1$

$\iiint_D (1)\,dz\,dy\,dx$ → Volume of the Region D.

$\iiint_D dy\,dx\,dz$, $\iiint_D dx\,dy\,dz$ → Order can be anything

Finding limits in triple Integral

→ In double Integral we used to sketch in 2D plane But here we will sketch in 3D.



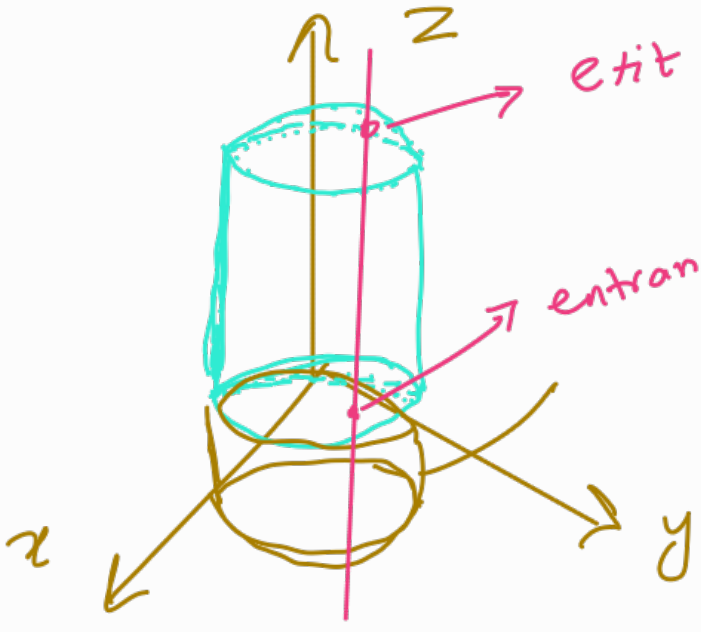
1) Sketch the Region D / Surface.

also Sketch the Shadow in the x-y plane

a) lets Say we are following order of Integration.

$$\iiint dz\,dy\,dx$$

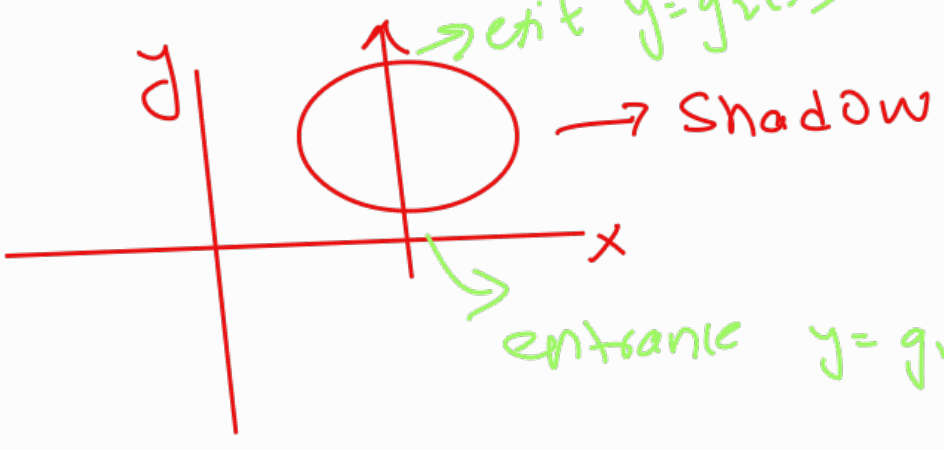
Making a line parallel to z-axis



exit $z = f_2(x,y)$ U.L

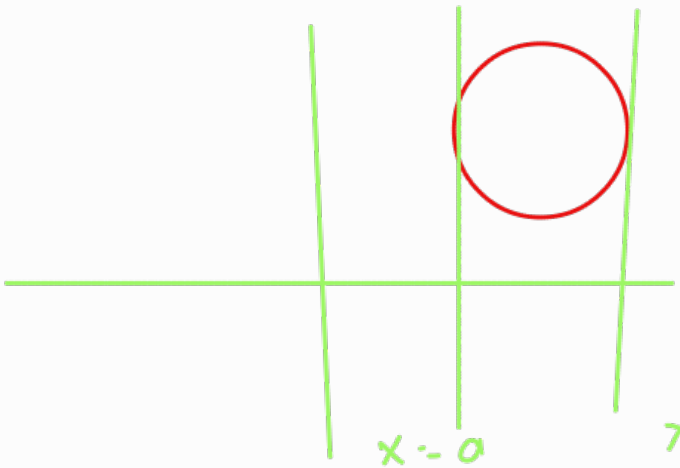
entrance point $z = f_1(x,y)$ L.L

Now make the shadow in x-y Plane. for finding x-y limits



Now finding the limit of y. Make a vertical line parallel to y-axis in x-y plane

Now the Last step is to find x-limits By making a vertical line in x-y Plane



$$\iiint_D f(x,y,z)\,dz\,dy\,dx = \int_a^b \iint_{g_1(x)}^{g_2(x)} \int_{f_1(x,y)}^{f_2(x,y)} f(x,y,z)\,dz\,dy\,dx$$

---

e.g find volume

$$\int_0^1 \int_x^1 \int_0^{y-x} dz\,dy\,dx$$

$$= \int_0^1 \int_x^1 \Big|_0^{y-x} dy\,dx$$

$$= \int_0^1 \int_x^1 (y-x)\,dy\,dx$$

$$\int_0^1 \int_x^1 \left| \frac{y^2}{2} - x \right| dx$$

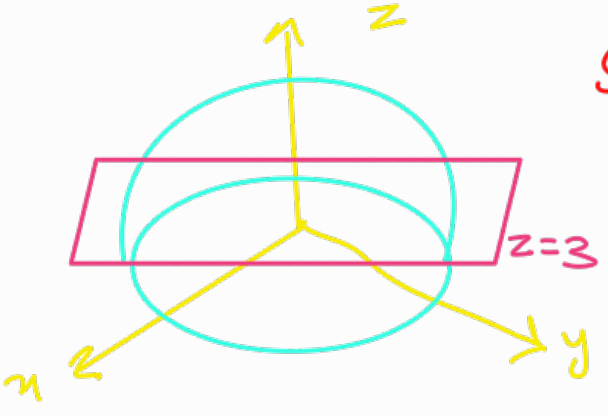
$$\int_0^1 \left[\frac{1}{2} y^2 - xy \right]_x^1$$

$$= \boxed{\frac{1}{6}} \quad \text{Ans}$$

---

e.g let S be sphere of radius 5 centered at the origin, and let D be the region under the sphere that lies above the plane $z=3$. Set up the limits of integration.

$$\iiint_R dz\,dy\,dx$$

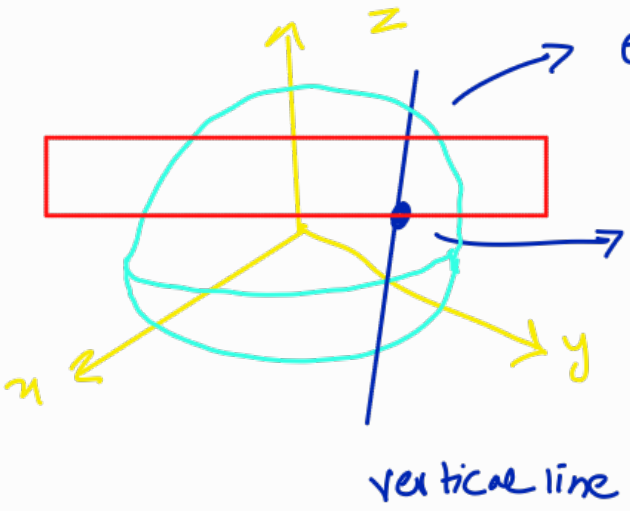


Sphere

$$x^2+y^2+z^2=(5)^2$$

finding z-limits

$z = 3$ (x-y plane)



vertical line

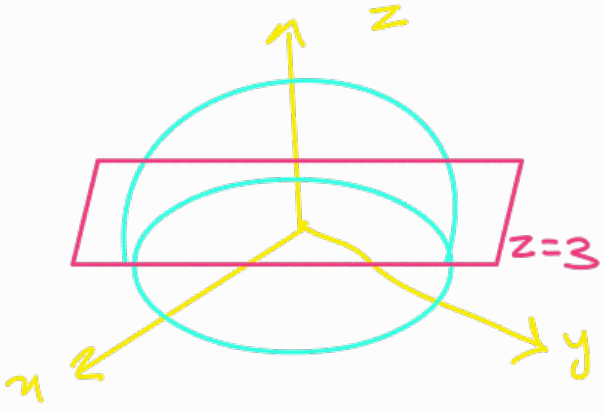
$x^2+y^2+z^2=25$

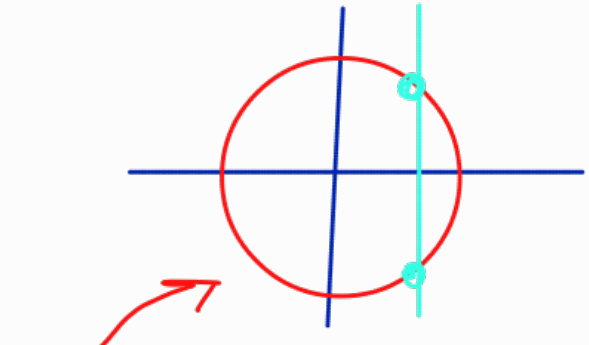
$z^2=25-x^2-y^2$

$z=\sqrt{25-x^2-y^2}$

$z=3$

finding y-limits of integration.





$z=3$ is in x-y plane

$x^2+y^2+(3)^2=25$

$x^2+y^2=16$

entrance

$y^2=16-x^2$

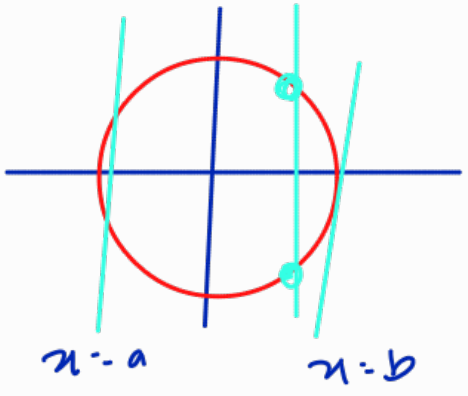
$y=\pm\sqrt{16-x^2}$

$$y=-\sqrt{16-x^2}$$

exit

$$y=+\sqrt{16-x^2}$$

finding x-limits



$x=-4$, $x=4$

$$\int_{-4}^{4}\int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}}\int_{z=3}^{\sqrt{25-x^2-y^2}} dz\,dy\,dx$$

→ final Answer.

(limits setup for tetrahedron)

Setup the limits of integration for the function $F(x, y, z)$ over the tetrahedron D whose vertices are $(0,0,0)$, $(1,1,0)$ $(0,1,0)$ and $(0,1,1)$ use the order of integratio

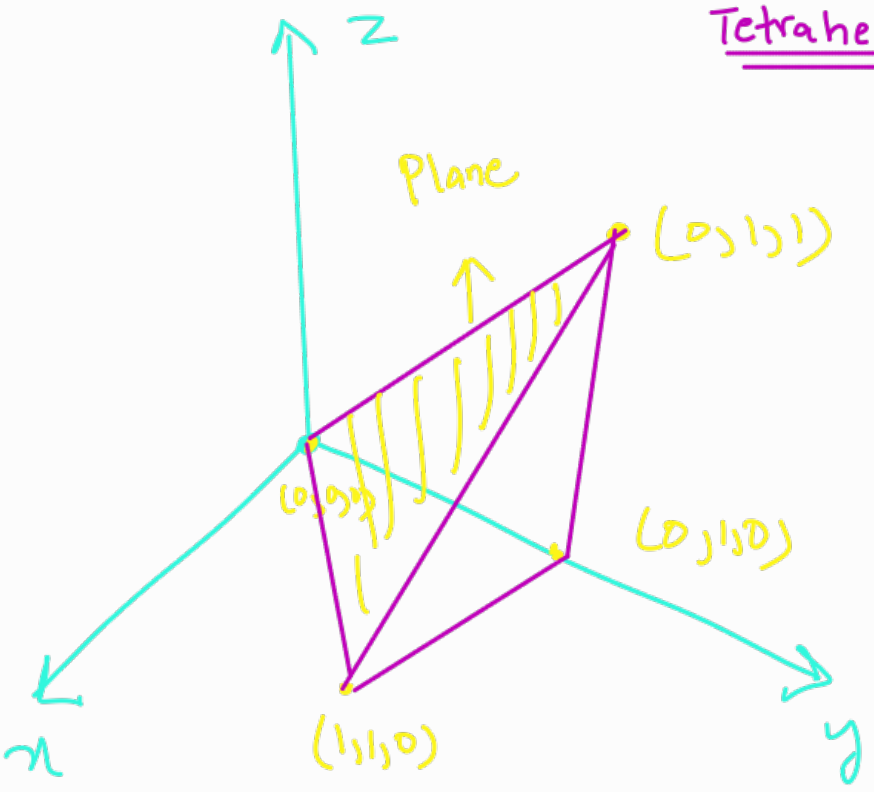
$$\iiint_D dz\,dy\,dx$$

Step 1
Sketch the surface in 3D.

Step 2
Make the equation.

Step 3 Set the limits up.

Tetrahedron



Equation of Plane

$ax + by + cz = d$

At $(0,0,0)$

$a(0) + b(0) + c(0) = d$

$\boxed{d = 0}$

At $(1,1,0)$

$a(1) + b(1) + c(0) = 0$

$\boxed{a + b = 0}$

$\boxed{a = -b}$

At $(0,1,1)$

$a(0) + b(1) + c(1) = 0$

$b + c = 0$

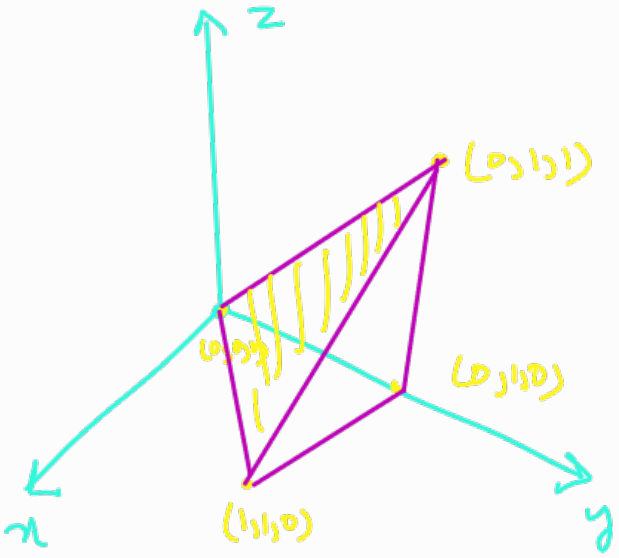
$\boxed{c = -b}$ $\boxed{c = a}$

hence $ax + by + cz = 0$

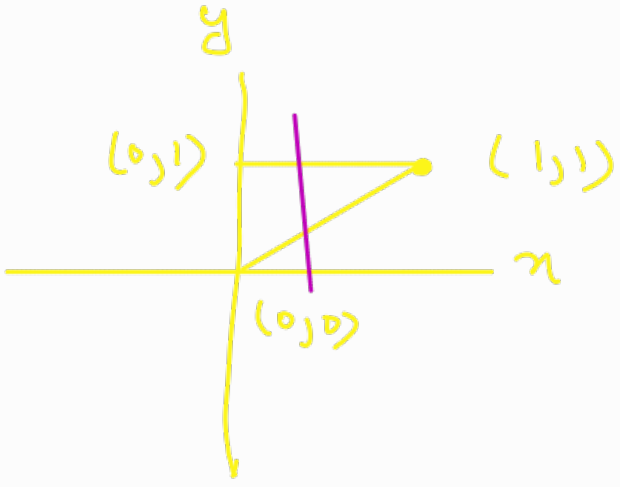
$ax - ay + az = 0$

$x - y + z = 0$

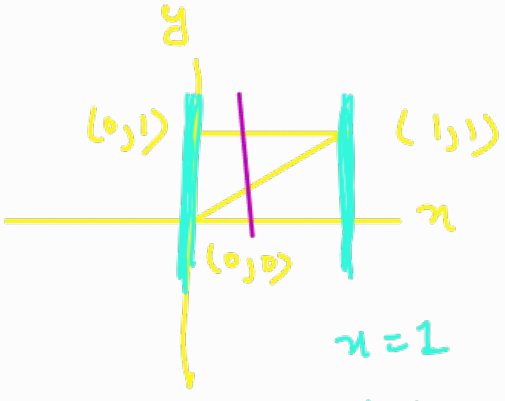
$\boxed{z = y - x}$



Shadow in x-y Plane



X-limits



$x=1$

U.L

$x=0$

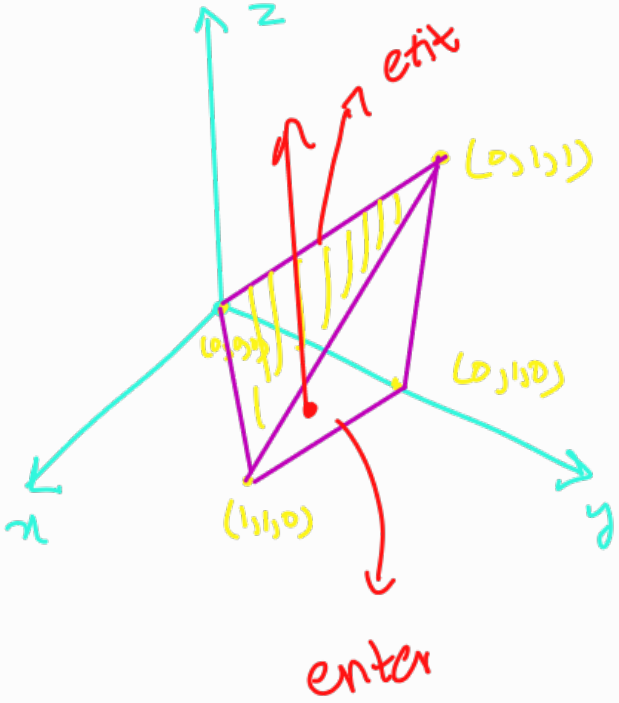
L.L

y-limits

$y=x \rightarrow$ L.L

$y=1 \rightarrow$ U.L

finding z- limits of Integration



Enter $z=0$

Exit $z=y-x$

final equation

$$\int_0^1 \int_x^1 \int_0^{y-x} dz\,dy\,dx$$



Ex 15.4

Q9 change cartesian integral into polar integral

$$\int_{-1}^{1}\int_{0}^{\sqrt{1-x^2}} dy\,dx \rightarrow \iint_D f(r,\theta)\, r\,dr\,d\theta$$

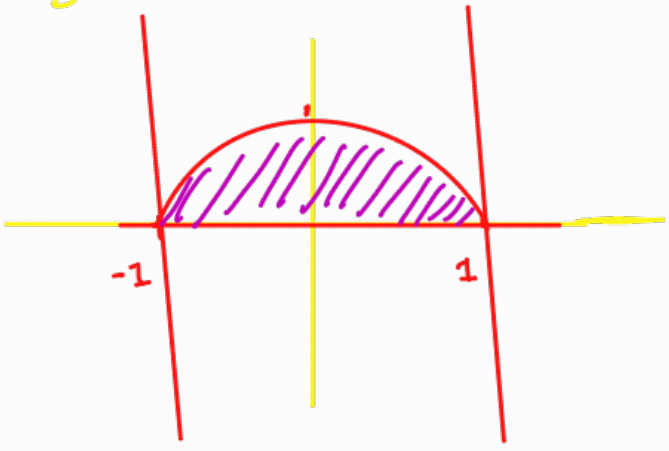
1) Sketching limits

$y=0$ $y=\sqrt{1-x^2}$

$x=1$

$x=-1$

x	-1	0	1
y	0	1	0



finding polar limits

$r=0$ and $r=$

$y=\sqrt{1-x^2}$

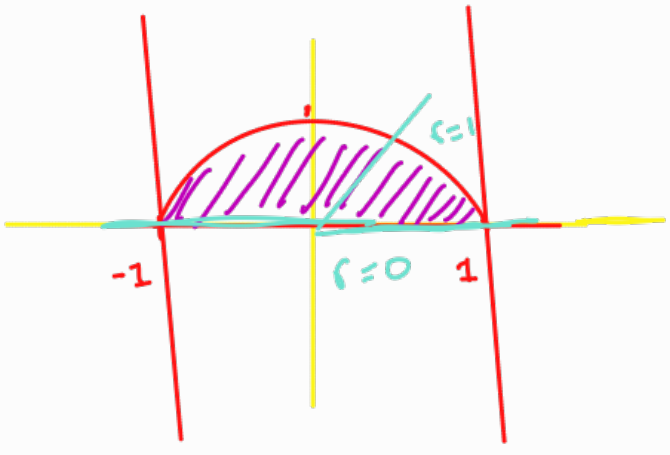
converting into polar

$y=\sqrt{1-x^2}$

$y^2=1-x^2$

$x^2+y^2=1$

$r=1$



$\theta = 0$ to π

$$\int_0^{\pi}\int_0^{1} f(r,\theta)\, r\,dr\,d\theta$$

$$\int_0^{\pi}\int_0^{1} \left|\frac{r^2}{2}\right| d\theta$$

$$\int_0^{\pi} \left(\frac{(1)^2}{2} - 0 \right) d\theta \rightarrow \int_0^{\pi} \frac{1}{2}\, d\theta$$

$$\rightarrow \left| \frac{\theta}{2} \right|_0^{\pi} \rightarrow \left(\frac{\pi}{2}\right) - (0) = \boxed{\frac{\pi}{2}}$$

Q10

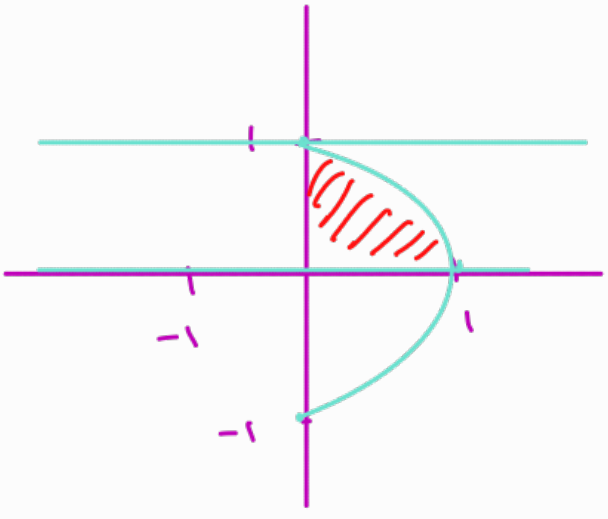
$$\int_0^1 \int_0^{\sqrt{1-y^2}} (x^2+y^2)\, dx\, dy \Rightarrow \iint_D f(r,\theta)\, r\, dr\, d\theta$$

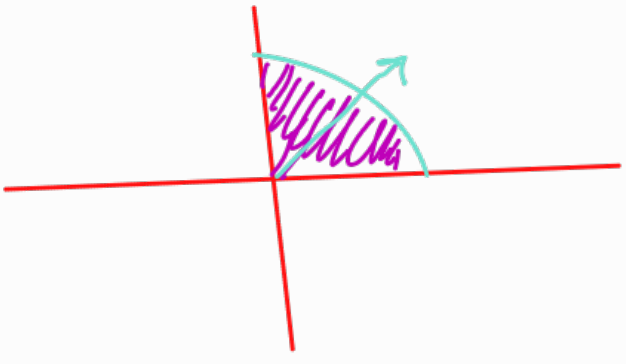
1) Sketch the region

$x = \sqrt{1-y^2}$ $\quad x = 0$

$y = 1$ $\quad y = 0$

y	-1	0	1
x	0	1	0





finding r-Limits of Integration.

$\boxed{r = 0}$

$x = \sqrt{1-y^2}$ $\quad \theta = 0$ to $\frac{\pi}{2}$

$x^2 = 1 - y^2$

$x^2 + y^2 = 1$

$\boxed{r = 1}$

$$\int_0^{\frac{\pi}{2}} \int_0^1 f(r,\theta)\, r\, dr\, d\theta$$

$$\int_0^{\frac{\pi}{2}} \int_0^1 r^2 \cdot r\, dr\, d\theta \rightarrow \int_0^{\frac{\pi}{2}} \int_0^1 r^3\, dr\, d\theta$$

$$\rightarrow \int_0^{\frac{\pi}{2}} \int_0^1 \frac{r^4}{4}\, d\theta \rightarrow \boxed{\frac{\pi}{8}}$$

Q12

$$\int_{-a}^{a} \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2+x^2}} dy\, dx$$

$y = \sqrt{a^2 + x^2}$

$\boxed{y^2 - x^2 = a^2}$

$-\sqrt{a^2 - x^2} = y$

$-a^2 + x^2 = y^2$

$\boxed{y^2 - x^2 = -a^2}$

Complete circle

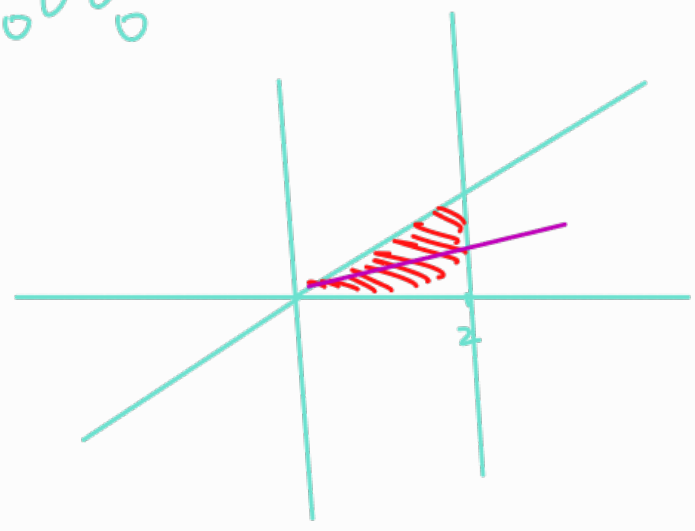
Q14

$$\int_0^2 \int_0^x y \, dy \, dx$$

1) Sketching the region.

$y = x$ $\quad y = 0$

$x = 2$ $\quad x = 0$



finding r-limits of integration

$r = 0$ to $r = ?$

$\theta =$ 0 to intersection of $x = 2$ $y = x$

$\tan^{-1}\left(\frac{2}{2}\right) = \frac{\pi}{4}$



V.L for r

$x = 2$

$r\cos\theta = 2$

$r = \frac{2}{\cos\theta}$

$r = 2\sec\theta$

$$\int_0^{\pi/4} \int_0^{2\sec\theta} f(r,\theta)\, r \, dr \, d\theta$$

$$\int_0^{\pi/4} \int_0^{2\sec\theta} r^2 \sin\theta \, dr \, d\theta$$

integrating r by keeping θ constant.

$$\int_0^{\pi/4} \int_0^{2\sec\theta} \frac{r^3}{3} \sin\theta \, d\theta$$

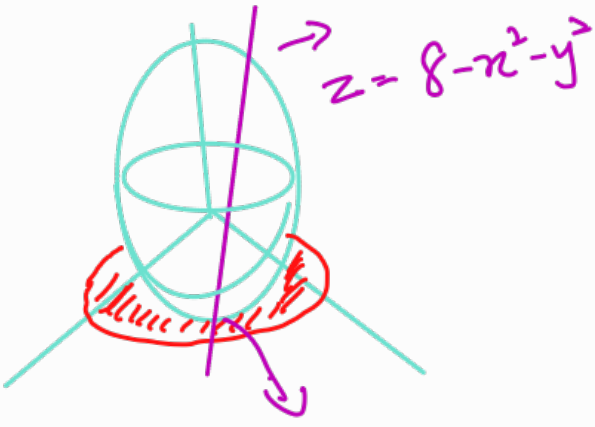
$$\int_0^{\pi/4} \left| \frac{8 \sec^3\theta \sin\theta}{3} \right| d\theta$$

$$\frac{8}{3} \int_0^{\pi/4} \left| \sec^3\theta \sin\theta \right| d\theta$$

Aggy I know smit

$\cap_{11}$

Ex 15.5 Let D be the region bounded by the paraboloids $z = 8 - x^2 - y^2$ and $z = x^2 + y^2$. Write six diff triple iterated integrals for the volume of D. Evaluate one of the integrals.



Paraboloid 1, Paraboloid 2

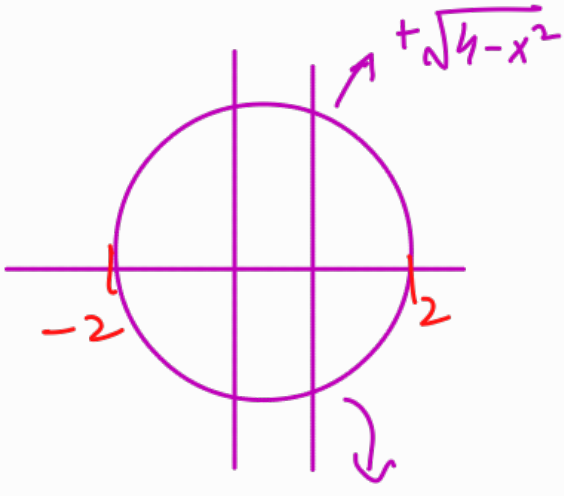
$z = 8 - x^2 - y^2$

$z = x^2 + y^2$

Intersection

$x^2 + y^2 = 8 - x^2 - y^2$

$\boxed{x^2 + y^2 = 4}$ → Shadow



Solving $\iiint dz\,dy\,dx$

$$2\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{+\sqrt{4-x^2}}\int_{z=x^2+y^2}^{8-x^2-y^2} dz\,dy\,dx$$

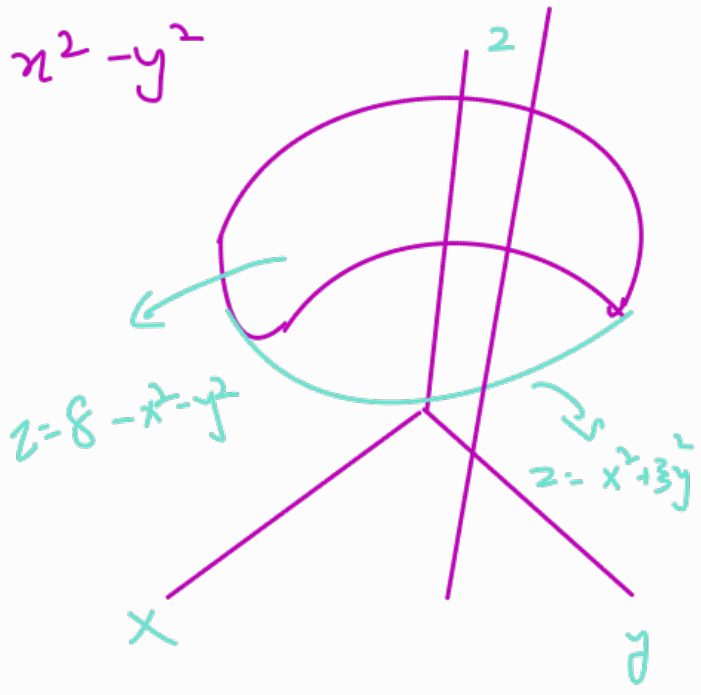
for x-limits

Put $y=0$ we get

$-2, 2$

Example 4

$z = x^2 + 3y^2$ $\qquad z = 8 - x^2 - y^2$



$8 - x^2 - y^2 = x^2 + 3y^2$

$8 = x^2 + x^2 + 3y^2 + y^2$

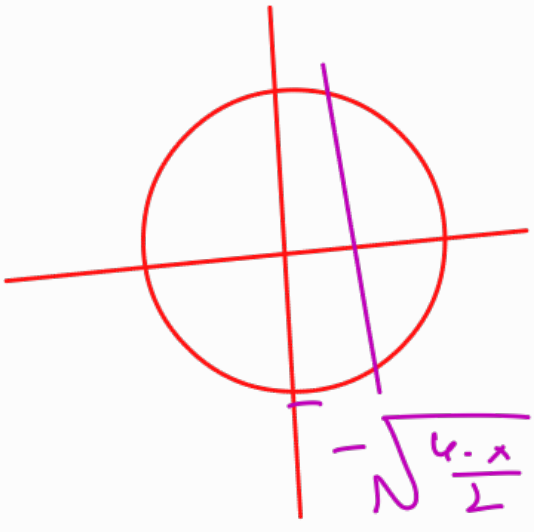
$8 = 2x^2 + 4y^2$

$8 = 2(x^2 + 2y^2)$

$$\boxed{x^2 + 2y^2 = 4}$$ → shadow

Solve $\iiint_D dz\,dy\,dx$

$$\int_{-2}^{2} \int_{-\sqrt{\frac{4-x}{2}}}^{+\sqrt{\frac{4-x}{2}}} \int_{x^2+3y^2}^{8-x^2-y^2} dz\,dy\,dx$$



$x^2 + 2y^2 = 4$

$2y^2 = 4 - x^2$

$y^2 = \frac{4-x^2}{2}$

$y = \pm\sqrt{\frac{4-x^2}{2}}$

$$\boxed{y = \pm\sqrt{\frac{4-x^2}{2}}}$$

Solving x Limits Since

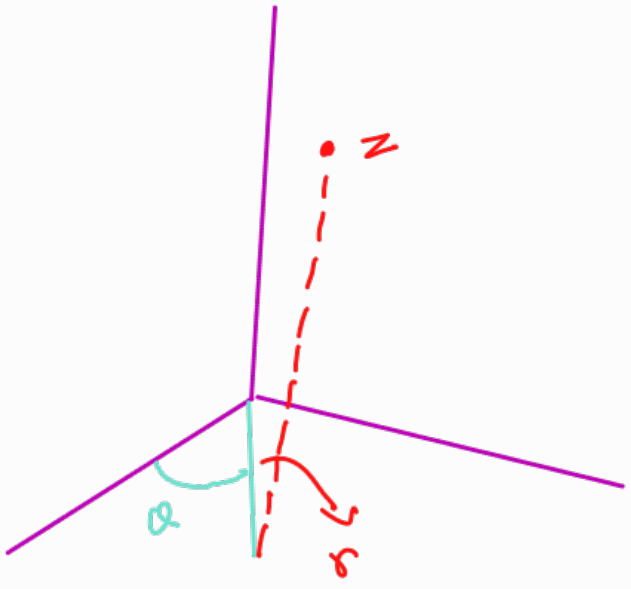
$y = 0$

$x^2 = 4$

$x = \pm 2$

Triple integrals in cylindrical coordinates

(x, y, z) is expressed in (r, θ, z)



$r \rightarrow$ distance out in the plane from $(0,0)$.

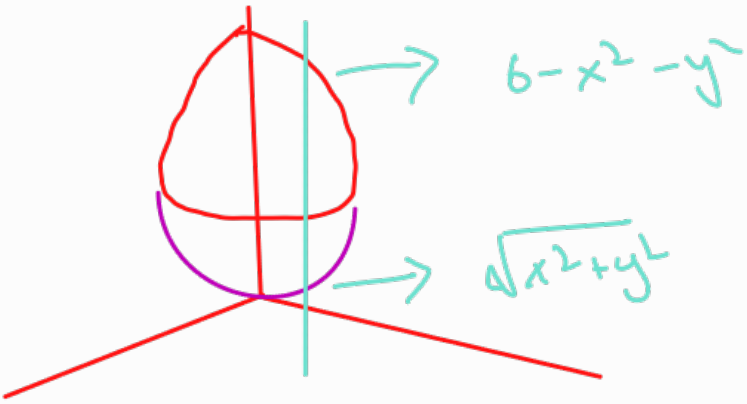
$z \rightarrow$ height.

$dx\,dy\,dz \rightarrow (r\,dr\,d\theta)\,dz$ or $dz\,r\,dr\,d\theta$

Volume

$$\iiint_D dz\,r\,dr\,d\theta$$

e.g. Find the volume b/w the cone $z = \sqrt{x^2+y^2}$ and the paraboloid $z = 6 - x^2 - y^2$



(finding z- limits)

U.L $z = 6 - x^2 - y^2$ converting to polar

$z = 6 - (x^2 + y^2)$

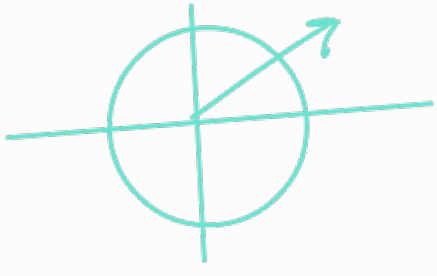
$\boxed{z = 6 - r^2}$

L.L $z = \sqrt{x^2 + y^2}$

$\boxed{z = r}$

$$\iiint_r^{6-r^2} dz\, r\, dr\, d\theta$$

Finding r - limits and θ- limits in x-y plane



To find circle eq. using intersection.

$z = 6 - x^2 - y^2$ $\quad z = \sqrt{x^2 + y^2}$

hence $\quad r = 6 - r^2$

$\boxed{r^2 + r - 6 = 0}$

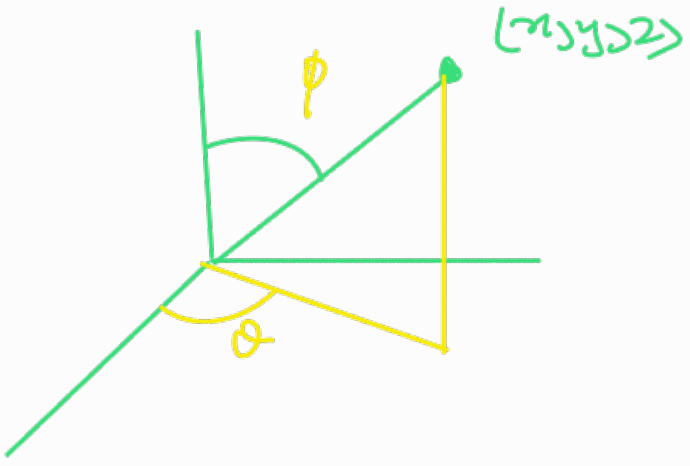
$\boxed{\begin{array}{l} r = -3 \\ r = 2 \end{array}}$

$$\int_0^{2\pi} \int_0^{2} \int_r^{6-r^2}$$

Spherical

(ρ, ϕ, θ)

- ρ → radius of Sphere
- ϕ → angle with the z-axis
- θ → angle with the x-axis



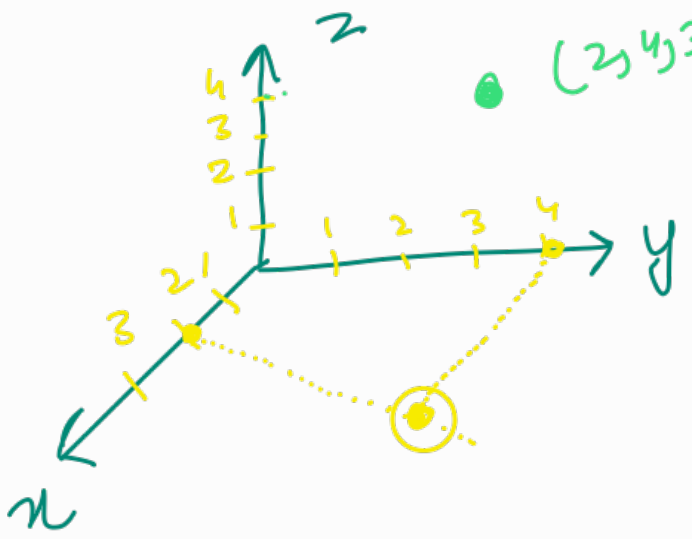
$$x = \rho \sin\phi \cos\theta$$
$$y = \rho \sin\phi \sin\theta$$
$$z = \rho \cos\phi$$

A point x, y, z lies on a Sphere

$$x = \rho \sin\phi \cos\phi$$
$$y = \rho \sin\phi \sin\phi$$
$$z = \rho \cos\phi$$

$$\int f(\rho, \phi, \theta)\, d\rho\, d\phi\, d\theta$$

Plotting points in 3- dimension System.



P(2, 4, 3)

x, y, z